

Theory of Manifold Interfaces

TMI-QG-2.0: A Minimal Predictive Model of Interface-Induced Geometric Decoherence

Working publication-oriented TEX version

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Abstract

This text assembles the minimal predictive version of the quantum-gravitational layer of the Theory of Manifold Interfaces. The model does not claim to be a completed theory of quantum gravity and does not derive the Standard Model. Its task is more modest and more precise: to formulate an effective interface-field mechanism of geometric decoherence, in which the distinguishability of geometric alternatives generates an additional contribution to the decoherence rate. The central prediction is

$$\Delta\Gamma_I = \Gamma_{\text{obs}} - \Gamma_{\text{std}} = \kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

Thus TMI-QG-2.0 defines a testable hypothesis: if an interface channel of geometric decoherence exists, the observed excess over standard decoherence should scale with the interface geometric contrast

$$x_I = |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

If such dependence is absent, the minimal version of the model is rejected or requires modification.

Contents

1	Status of the version	1
2	Axiomatic core of the minimal model	2
2.1	Axiom of geometric admissibility	2
2.2	Axiom of quantum distribution of possibilities	2
2.3	Axiom of interface structuring	2
3	The minimal two-geometry model	3
3.1	Two geometric alternatives	3
3.2	Interface-induced decoherence	3
3.3	Degree of structuring	3

4 Action of the interface field	4
4.1 Minimal system	4
4.2 Interface-field Lagrangian	4
4.3 Source of the interface field	4
5 Interface distance between geometries	5
5.1 General effective definition	5
5.2 Minisuperspace toy model	5
6 Main prediction of the model	5
7 Numerical example	6
8 Comparison with standard decoherence	6
9 Observable quantity and hypotheses	7
10 Synthetic verification dataset	7
11 Statistical test form	8
12 Domain of validity	9
12.1 What the model already provides	9
12.2 What the model does not yet prove	9
12.3 Phenomenological parameters	9
12.4 The measure problem	9
13 Final formulation of TMI-QG-2.0	10
14 Short publication abstract	10
15 Next stage	11

1 Status of the version

TMI-QG-2.0 is not a final Theory of Everything and not a completed theory of quantum gravity. It is a minimal effective model of interface-induced geometric decoherence.

The text fixes the transition from the general architecture of the Theory of Manifold Interfaces to its first testable model. The central physical chain is

$$g \Rightarrow \text{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \text{Dist}_{\text{poss}}(\Psi_I; g) \Rightarrow \text{Struct}_I.$$

Its meaning is

geometry defines admissibility;

the quantum state defines the distribution of possibilities;

the interface transforms the distribution of possibilities into structure.

In version 2.0 this chain is compressed into a predictive model:

$$(a_1, a_2, c_1, c_2, \kappa_I) \Rightarrow \Gamma_I \Rightarrow C_{\text{TMI}}(t) \Rightarrow \text{Struct}_I(t) \Rightarrow \text{statistical test}.$$

2 Axiomatic core of the minimal model

2.1 Axiom of geometric admissibility

Let $g_{\mu\nu}$ be a metric on spacetime $\mathcal{M}_U$. The metric defines the causal structure and the set of admissible causal transitions:

$$g \Rightarrow \mathcal{C}_{I,g}^+(x) \Rightarrow J_g^+(x) \Rightarrow \text{Adm}_c(g).$$

Here

$$\mathcal{C}_{I,g}^+(x) := \{v \in T_x \mathcal{M}_U \setminus \{0\} \mid g_x(v, v) \leq 0, \text{ } v \text{ is future-directed}\}$$

is the interface representation of the future causal cone. Its boundary coincides with the light cone of standard Lorentzian geometry:

$$\partial \mathcal{C}_{I,g}^+(x) \equiv \mathcal{C}_{\text{light},g}^+(x).$$

2.2 Axiom of quantum distribution of possibilities

For a fixed geometry g , the interface space of possibilities is defined as

$$\mathcal{H}_I(g) = L^2(\text{Adm}_c(g), \mathcal{B}_g, \mu_g),$$

where $\mathcal{B}_g$ is a chosen σ -algebra and μ_g is an effective or regularized measure on the space of admissible transitions. A quantum state

$$\Psi_I \in \mathcal{H}_I(g)$$

defines a distribution of possibilities:

$$d\mathbb{P}_{\Psi,g}(\Phi) = |\psi_I(\Phi)|^2 d\mu_g(\Phi).$$

2.3 Axiom of interface structuring

The interface transforms a distribution of possibilities into structure:

$$I_Q : \text{Dist}_{\text{poss}}(\Psi_I; g) \longrightarrow \text{Struct}_I.$$

Operationally this may be written as a quantum channel:

$$\mathcal{E}_I(\rho) = \sum_k K_k^{(I)} \rho K_k^{(I)\dagger}, \quad \sum_k K_k^{(I)\dagger} K_k^{(I)} = \mathbf{1}.$$

In the limiting case of complete reduction,

$$\rho'_I \approx \sum_k p_k |s_k\rangle \langle s_k|, \quad p_k = \text{Tr}(E_k^{(I)} \rho), \quad E_k^{(I)} = K_k^{(I)\dagger} K_k^{(I)}.$$

3 The minimal two-geometry model

3.1 Two geometric alternatives

Let there be two admissible geometric configurations:

$$|g_1\rangle, \quad |g_2\rangle.$$

They form the minimal space of alternatives:

$$\mathcal{H}_g^{(2)} = \text{span}\{|g_1\rangle, |g_2\rangle\}.$$

For the first model we assume orthonormality:

$$\langle g_i | g_j \rangle = \delta_{ij}.$$

The quantum state of geometry is

$$|\Psi_g\rangle = c_1|g_1\rangle + c_2|g_2\rangle, \quad |c_1|^2 + |c_2|^2 = 1.$$

The density matrix is

$$\rho_g(0) = |\Psi_g\rangle\langle\Psi_g| = \begin{pmatrix} |c_1|^2 & c_1\overline{c_2} \\ c_2\overline{c_1} & |c_2|^2 \end{pmatrix}.$$

The diagonal elements define probabilities, while the off-diagonal elements ρ_{12} and ρ_{21} define geometric coherence.

3.2 Interface-induced decoherence

Introduce the coherence-preservation coefficient $\eta_I(t)$:

$$\rho_{12}(t) = \eta_I(t)\rho_{12}(0).$$

If $0 \leq \eta_I(t) \leq 1$, then

$$\rho_g(t) = \begin{pmatrix} |c_1|^2 & \eta_I(t)c_1\overline{c_2} \\ \eta_I(t)c_2\overline{c_1} & |c_2|^2 \end{pmatrix}.$$

The interface-induced decoherence rate is $\Gamma_I(t) \geq 0$:

$$\eta_I(t) = \exp\left(-\int_0^t \Gamma_I(s)ds\right).$$

For a constant rate,

$$\eta_I(t) = e^{-\Gamma_I t}, \quad |\rho_{12}(t)| = e^{-\Gamma_I t}|\rho_{12}(0)|.$$

3.3 Degree of structuring

The minimal degree of structuring is

$$\text{Struct}_I(t) := 1 - \frac{|\rho_{12}(t)|}{|\rho_{12}(0)|}.$$

Therefore,

$$\text{Struct}_I(t) = 1 - \eta_I(t),$$

and, for a constant rate,

$$\boxed{\text{Struct}_I(t) = 1 - e^{-\Gamma_I t}.$$

4 Action of the interface field

4.1 Minimal system

The minimal physical system is

$$(g_{\mu\nu}, \Psi_g, \chi_I),$$

where χ_I is an effective interface field. The total action is

$$S_{\text{TMI}} = S_{\text{GR}}[g] + S_Q[\Psi_g, g] + S_I[\chi_I, g] + S_{\text{int}}[\chi_I, \Psi_g, g].$$

The gravitational part is

$$S_{\text{GR}}[g] = \frac{c^4}{16\pi G} \int_{\mathcal{M}_U} R \sqrt{-g} d^4x.$$

4.2 Interface-field Lagrangian

In the minimal version $\chi_I : \mathcal{M}_U \rightarrow \mathbb{R}$ is a real scalar field. Its Lagrangian is

$$\mathcal{L}_\chi = -\frac{1}{2}g^{\mu\nu}\nabla_\mu\chi_I\nabla_\nu\chi_I - \frac{1}{2}m_I^2\chi_I^2 - \frac{\beta_I}{4}\chi_I^4 - \frac{1}{2}\xi_IR\chi_I^2 + J_I\chi_I.$$

Here m_I is the scale of the interface field, β_I is self-interaction, ξ_I is curvature coupling, and J_I is the source of the interface field.

The equation of motion is

$$\boxed{\square_g\chi_I - (m_I^2 + \xi_IR)\chi_I - \beta_I\chi_I^3 = -J_I.}$$

4.3 Source of the interface field

The interface contrast for two geometries is defined in a soft normalized form:

$$D_g(\Psi_g) = |c_1||c_2|d(g_1, g_2).$$

The source is

$$J_I = \alpha_I D_g(\Psi_g).$$

In the stationary weak-field approximation,

$$\nabla_\mu\chi_I \approx 0, \quad R \approx 0, \quad \beta_I \approx 0.$$

Then

$$\chi_I = \frac{J_I}{m_I^2} = \frac{\alpha_I}{m_I^2} D_g(\Psi_g).$$

If

$$\Gamma_I = \lambda_I \chi_I^2,$$

then

$$\boxed{\Gamma_I = \frac{\lambda_I \alpha_I^2}{m_I^4} |c_1|^2 |c_2|^2 d_I^2(g_1, g_2).}$$

5 Interface distance between geometries

5.1 General effective definition

Strictly speaking, one should compare not coordinate representations of metrics but equivalence classes of geometries:

$$[g] = \{\varphi^* g \mid \varphi \in \text{Diff}(\mathcal{M}_U)\}.$$

The effective interface distance is

$$d_I([g_1], [g_2]) = \frac{1}{\mathcal{N}_I} \inf_{\varphi \in \text{Diff}(\mathcal{M}_U)} \left[\int_{\Omega} w_I(x) \|g_1 - \varphi^* g_2\|_{g_0}^2 dV_{g_0} \right]^{1/2}.$$

Here $w_I(x) \geq 0$ is an interface weight, g_0 is a regularizing background structure, and $\mathcal{N}_I$ is a normalization factor.

5.2 Minisuperspace toy model

For the first computable case, geometry is represented by the scale factor a :

$$g_1 = g(a_1), \quad g_2 = g(a_2).$$

Define

$$d_I(g(a_1), g(a_2)) = \frac{|\ln(a_1/a_2)|}{\sigma_a}.$$

Then

$$\Gamma_I = \frac{\lambda_I \alpha_I^2}{m_I^4 \sigma_a^2} |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

Let

$$\kappa_I = \frac{\lambda_I \alpha_I^2}{m_I^4 \sigma_a^2}.$$

The central rate of the model becomes

$$\Gamma_I = \kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

6 Main prediction of the model

The normalized coherence is

$$C_I(t) = \frac{|\rho_{12}(t)|}{|\rho_{12}(0)|}.$$

Thus

$$C_I(t) = \exp \left[-\kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right) t \right].$$

For the maximal superposition $|c_1|^2 = |c_2|^2 = 1/2$,

$$\Gamma_I = \frac{\kappa_I}{4} \ln^2 \left(\frac{a_1}{a_2} \right), \quad C_I(t) = \exp \left[-\frac{\kappa_I}{4} \ln^2 \left(\frac{a_1}{a_2} \right) t \right].$$

The characteristic interface-decoherence time is

$$\tau_I = \frac{1}{\Gamma_I} = \frac{1}{\kappa_I |c_1|^2 |c_2|^2 \ln^2(a_1/a_2)}.$$

7 Numerical example

Take a maximal superposition and $\kappa_I = 1$. Then

$$\Gamma_I = \frac{1}{4} \ln^2(a_1/a_2).$$

a_1/a_2	$\ln(a_1/a_2)$	Γ_I	$\tau_I = 1/\Gamma_I$
2	0.693	0.120	8.32
10	2.303	1.326	0.754
100	4.605	5.302	0.189

For $C_I(t) = e^{-\Gamma_I t}$:

t	$a_1/a_2 = 2$	$a_1/a_2 = 10$	$a_1/a_2 = 100$
0	1.000	1.000	1.000
0.5	0.942	0.515	0.071
1	0.887	0.265	0.005
2	0.786	0.071	0.000025
5	0.548	0.0013	≈ 0

The conclusion is

interface-induced decoherence grows as the square of the logarithm of the ratio of ge

8 Comparison with standard decoherence

Let standard decoherence have rate Γ_{std} :

$$C_{\text{std}}(t) = e^{-\Gamma_{\text{std}} t}.$$

TMI adds an interface rate:

$$\Gamma_{\text{total}} = \Gamma_{\text{std}} + \Gamma_I.$$

Therefore,

$$C_{\text{TMI}}(t) = e^{-(\Gamma_{\text{std}} + \Gamma_I)t}.$$

The relative difference is

$$\frac{C_{\text{TMI}}(t)}{C_{\text{std}}(t)} = e^{-\Gamma_I t}.$$

The excess suppression of coherence is

$$\Delta_I(t) = 1 - e^{-\Gamma_I t}.$$

In the minisuperspace model,

$$\Delta_I(t) = 1 - \exp \left[-\kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right) t \right].$$

9 Observable quantity and hypotheses

The main observable excess is

$$\Delta\Gamma_I = \Gamma_{\text{obs}} - \Gamma_{\text{std}}.$$

The interface parameter of geometric contrast is

$$x_I := |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

Therefore, the TMI-QG prediction is

$$\Delta\Gamma_I = \kappa_I x_I.$$

The null hypothesis is

$$H_0 : \kappa_I = 0.$$

The TMI hypothesis is

$$H_{\text{TMI}} : \kappa_I > 0.$$

Equivalently,

$$H_{\text{TMI}} : \Gamma_{\text{obs}} = \Gamma_{\text{std}} + \kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

The minimal version is falsified if, with sufficient precision,

$$\Gamma_{\text{obs}} - \Gamma_{\text{std}} = 0$$

for significant $a_1/a_2 \neq 1$, or if the dependence does not scale as x_I .

10 Synthetic verification dataset

Take

$$|c_1|^2 = |c_2|^2 = \frac{1}{2}, \quad \kappa_I = 0.5, \quad \Gamma_{\text{std}} = 0.1.$$

Then

$$x_I = \frac{1}{4} \ln^2(a_1/a_2), \quad \Delta\Gamma_I = 0.5x_I, \quad \Gamma_{\text{obs}} = 0.1 + 0.5x_I.$$

a_1/a_2	$\ln(a_1/a_2)$	x_I	$\Delta\Gamma_I$	Γ_{obs}
1	0.000	0.000	0.000	0.100
2	0.693	0.120	0.060	0.160
5	1.609	0.648	0.324	0.424
10	2.303	1.325	0.663	0.763
50	3.912	3.826	1.913	2.013
100	4.605	5.302	2.651	2.751

Coherence at $t = 1$:

a_1/a_2	$C_{\text{std}}(1)$	$C_{\text{TMI}}(1)$	$\Delta_I(1)$
1	0.905	0.905	0.000
2	0.905	0.852	0.058
5	0.905	0.655	0.277
10	0.905	0.466	0.485
50	0.905	0.134	0.852
100	0.905	0.064	0.929

11 Statistical test form

For scenarios $i = 1, \dots, n$, define

$$x_I^{(i)} = |c_1^{(i)}|^2 |c_2^{(i)}|^2 \ln^2 \left(\frac{a_1^{(i)}}{a_2^{(i)}} \right),$$

$$y_i = \Delta\Gamma^{(i)} = \Gamma_{\text{obs}}^{(i)} - \Gamma_{\text{std}}^{(i)}.$$

The minimal statistical model is

$$y_i = \kappa_I x_I^{(i)} + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0.$$

If the errors have variances σ_i^2 , use weights

$$w_i = \frac{1}{\sigma_i^2}.$$

The estimator is

$$\hat{\kappa}_I = \frac{\sum_{i=1}^n w_i x_I^{(i)} y_i}{\sum_{i=1}^n w_i (x_I^{(i)})^2}.$$

The standard error is

$$\text{SE}(\hat{\kappa}_I) = \left(\sum_{i=1}^n w_i (x_I^{(i)})^2 \right)^{-1/2}.$$

The criterion for support is

$$\hat{\kappa}_I > 0 \quad \text{and} \quad 0 \notin \text{CI}(\hat{\kappa}_I).$$

For an approximate 95% criterion,

$$\hat{\kappa}_I - 1.96 \text{ SE}(\hat{\kappa}_I) > 0.$$

12 Domain of validity

12.1 What the model already provides

The model already provides:

1. a computable quantity Γ_I ;
2. the coherence-suppression law $|\rho_{12}(t)| = e^{-\Gamma_I t} |\rho_{12}(0)|$;
3. a distinction from standard decoherence;
4. a falsifiable hypothesis $\Delta\Gamma_I \propto x_I$;
5. a statistical test form.

12.2 What the model does not yet prove

The model does not prove:

1. a complete theory of quantum gravity;
2. a derivation of the Standard Model $SU(3) \times SU(2) \times U(1)$;
3. the fundamental origin of all masses, charges, and particle generations;
4. a full measure μ_G on the space of geometries;
5. a general distance formula between arbitrary geometries;
6. a universal mechanism of wave-function collapse.

12.3 Phenomenological parameters

The coefficient κ_I is an effective parameter in the current version:

κ_I has not yet been derived from fundamental constants.

A stricter version should obtain

$$\kappa_I = f(\lambda_I, \alpha_I, m_I, \sigma_a, \dots)$$

or more deeply,

$$\kappa_I = f(S_I, \chi_I, g, \Psi).$$

12.4 The measure problem

The measure μ_G on the space of geometries is assumed to be effective or regularized in the current version. Possible future choices include:

1. a discrete measure;
2. a minisuperspace measure;
3. a cylindrical measure;
4. an effective path-integral measure.

13 Final formulation of TMI-QG-2.0

The minimal TMI-QG model predicts an additional geometry-dependent contribution to the decoherence rate, generated by interface distinguishability of geometric alternatives.

Main formula:

$$\Delta\Gamma_I = \Gamma_{\text{obs}} - \Gamma_{\text{std}} = \kappa_I |c_1|^2 |c_2|^2 \ln^2 \left(\frac{a_1}{a_2} \right).$$

Main testable dependence:

$$\Delta\Gamma_I = \kappa_I x_I, \quad x_I = |c_1|^2 |c_2|^2 \ln^2(a_1/a_2).$$

Main statistical hypothesis:

$$H_0 : \kappa_I = 0, \quad H_{\text{TMI}} : \kappa_I > 0.$$

Correct scientific status:

TMI-QG proposes an interface-field mechanism for geometric decoherence.

Incorrect status:

TMI-QG should not be presented as a completed quantum gravity or Theory of Everything.

14 Short publication abstract

The minimal version of TMI-QG formulates an effective interface-field mechanism of geometric decoherence. In the two-geometry minisuperspace approximation, quantum geometry is represented by the state

$$|\Psi_g\rangle = c_1 |g(a_1)\rangle + c_2 |g(a_2)\rangle.$$

The interface geometric contrast is defined as

$$x_I = |c_1|^2 |c_2|^2 \ln^2(a_1/a_2).$$

The model predicts an additional contribution to the decoherence rate:

$$\Delta\Gamma_I = \kappa_I x_I.$$

Therefore,

$$C_{\text{TMI}}(t) = \exp[-(\Gamma_{\text{std}} + \kappa_I x_I)t].$$

The model is falsifiable: if the observed excess $\Gamma_{\text{obs}} - \Gamma_{\text{std}}$ does not show a positive linear dependence on x_I , the minimal version of the interface channel of geometric decoherence is rejected or requires modification.

15 Next stage

After TMI-QG-2.0, the natural next steps are:

1. construct a stricter measure μ_G , or explicitly fix the regularization;
2. move from two geometries to an N -geometry and continuous model;
3. derive κ_I from a more fundamental action $S_I[\chi_I, g, \Psi]$;
4. build a numerical toy model with a concrete potential $U_{\text{mini}}(a)$;
5. compare predictions with Wheeler-DeWitt, standard decoherence, path-integral approaches, semiclassical gravity, and LQG;
6. separate the physical TMI-QG branch from the mathematical branch of the category of interfaces.